

BABU BANARSI DAS UNIVERSITY

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CASE STUDY ON

Social Media Impact on Teen mental health

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Problem Statement

The rapid increase in social media usage among teenagers has raised concerns regarding its effect on mental health. Factors such as excessive screen time, stress levels, reduced sleep hours, and social media addiction may contribute to depression among teens.

The objective of this project is to analyze a Teen Mental Health dataset using Hadoop MapReduce and identify patterns related to depression. The project processes large-scale data using distributed computing techniques and generates analytical results.

Objective

The objectives of this project are:

1. To store the dataset in Hadoop Distributed File System (HDFS).
2. To process the dataset using MapReduce.
3. To count depressed and non-depressed students.
4. To analyze mental health trends using distributed data processing.

Dataset Description

The dataset contains attributes related to:

- Student ID
- Social Media Usage Hours
- Stress Level
- Sleep Hours
- Depression Label
- Anxiety Level
- Addiction Level

The attribute **Depression Label** is used as the key analytical field.

Methodology

Step 1: Data Upload to HDFS

The dataset was uploaded into Hadoop Distributed File System using:

```
cloudera@quickstart project]$ hdfs dfs -put /media/sf_BigDataProject/Teen_Mental_Health_Dataset.csv /mentalhealth/
```

Step 2: Mapper Function

The mapper reads each record from the dataset, extracts the depression label, and emits

Step 3: Reducer Function

The reducer groups records by depression label and sums all counts.

MapReduce Algorithm

Mapper Algorithm:

1. Read each row from CSV file
2. Extract depression label
3. Emit (depression_label,1)

Reducer Algorithm:

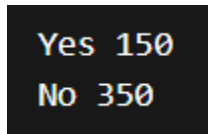
1. Receive grouped key-value pairs
2. Sum all values for each key
3. Output (key,total count)

Execution Command

```
hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \  
-files mapper.py, reducer.py \  
-mapper "python mapper.py" \  
-reducer "python reducer.py" \  
-input /mentalhealth/Teen_Mental_Health_Dataset.csv \  
-output /output1
```

Output

The final output generated by Hadoop MapReduce was:



```
Yes 150
No 350
```

Interpretation:

- 150 students were classified as depressed.
- 350 students were classified as non-depressed.

Result Analysis

The analysis shows that Hadoop successfully processed the teen mental health dataset and generated depression-related statistics.

The MapReduce model efficiently counted records and demonstrated distributed data processing on a large dataset.

Conclusion

This project demonstrated the implementation of Hadoop and MapReduce for analyzing teen mental health data.

The dataset was stored in HDFS, processed using mapper and reducer scripts, and the final output identified depression trends among students.

The project shows that Big Data technologies can be used for analyzing real-world mental health datasets efficiently.

Future Scope

This project can be extended further using:

- Hive for SQL-based analysis
- Pig for data transformation
- Machine Learning for depression prediction
- Visualization dashboards for mental health insights